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Enhanced Diffusion and Chemotaxis of Enzymes

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Keywords

Fokker-Planck equation, microfluidics, fluorescence correlation spectroscopy, FCS, urease, catalase, aldolase

Abstract

Many enzymes appear to diffuse faster in the presence of substrate and to drift either up or down a concentration gradient of their substrate. Observations of these phenomena, termed enhanced enzyme diffusion (EED) and enzyme chemotaxis, respectively, lead to a novel view of enzymes as active matter. Enzyme chemotaxis and EED may be important in biology and could have practical applications in biotechnology and nanotechnology. They are also of considerable biophysical interest; indeed, their physical mechanisms are still quite uncertain. This review provides an analytic summary of experimental studies of these phenomena and of the mechanisms that have been proposed to explain them and offers a perspective on future directions for the field.

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1. INTRODUCTION

Studies over the past decade suggest that nonmotor enzymes engage in active, translational motion. Two phenomena have been noted. One, termed enhanced enzyme diffusion (EED), is an increase in the enzymes' diffusion constant induced, typically, by provision of substrate. The other, enzyme chemotaxis, is a tendency for enzymes to move up or down a substrate concentration gradient. These phenomena have been seen for both fast exothermic enzymes like urease and slow endothermic enzymes like aldolase. These results open a new perspective of enzymes as active matter (27) and have potential practical and biological implications. For example, enzyme chemotaxis can be used to separate catalytically active and inactive enzymes (12) and might contribute to the assembly of intracellular metabolons (69, 75) and to intracellular signaling (65). Several mechanisms have been proposed for these phenomena, but their physical basis is still a subject of active research.

In this review, we analyze relationships among the various experimental and theoretical studies and discuss general theoretical considerations and frameworks for this field. [Related reviews also summarize recent work on EED and chemotaxis (1, 4, 13, 73).] First for EED and then for chemotaxis, we review experimental findings and analyze molecular mechanisms that have been proposed to account for these findings. Central questions include how to explain observations of EED and chemotaxis in general, whether—or when—EED is an active process, and how to reconcile apparently inconsistent experimental observations. We conclude with a summary of key points and potential directions for future work to further elucidate these intriguing phenomena.

Enhanced enzyme diffusion (EED): an apparent speedup of translational enzyme diffusion, usually reported as an increase in the diffusion coefficient

Enzyme chemotaxis: apparent drift of an enzyme up (attractive) or down (repulsive) a chemical gradient

Active mechanism: proposed mechanism of EED or chemotaxis is active if it requires the release of chemical energy by enzyme catalysis

2. EXPERIMENTAL STUDIES OF ENHANCED ENZYME DIFFUSION

This section summarizes experimental reports of increases in enzyme diffusion coefficients, focusing on enzymes for which an active mechanism, such as execution of swimming motions by the

Table 1 Turnover rates, hydrodynamic radii, minimum thrust speeds, and required reaction free energies of enzymes reported to show enhanced enzyme diffusion

Enzyme	Turnover rate (s^{-1})	Radius (nm)	v_{min} ($m.s^{-1}$)	$-\Delta G^{\circ}_{req}$ ($kJ.mol^{-1}$)
T4 DNA polymerase	0.5 (7)	4.6 (9)	1×10^{-2}	1×10^{7}
Aldolase	5 (25)	4.9 (11)	9×10^{-3}	8×10^{5}
T7 RNA polymerase	4 (39)	6.0[a]	6×10^{-3}	6×10^{5}
Hexokinase	300 (75)	6.3 (38)	5×10^{-3}	6,000
ATP synthase	1,000 (23)	6.6 (8)	5×10^{-3}	2,000
Alkaline phosphatase	3,000 (46)	7.7 (15)	3×10^{-3}	400
Catalase	10,000 (46)	5.3 (11)	7×10^{-3}	300
Urease	10,000 (46)	7.0 (18)	4×10^{-3}	100
Acetylcholinesterase	20,000 (19)	8.8 (47)	3×10^{-3}	40

[a]Calculated with HYDROPRO (44) from Protein Data Bank structure 4RNP (58).
For details, see Reference 17. Citations for individual cells are in parentheses.

enzyme, has been proposed. We also discuss cases in which EED was expected but not observed and consider potential experimental artifacts.

2.1. Positive Observations

Apparent EED has been reported for various enzymes, including ATPase (8), T7 RNA polymerase (71), T4 DNA polymerase (55), hexokinase (75), aldolase (75), alkaline phosphatase (46), acetylcholinesterase (28), jack bean urease (27, 28, 43, 46, 54, 70), and catalase (29, 46, 54). The relative diffusion enhancements, measured in homogeneous solution at the highest tested substrate concentrations, range from 15% to 80% (**Table 1**). Although these increases are similar, the turnover rates of these enzymes span orders of magnitude (**Table 1**). This discrepancy argues against an active mechanism as a general explanation of EED. Most of these studies used fluorescence correlation spectroscopy (FCS), which measures enzyme diffusion rates in homogeneous solution. However, one (71) used fluorescence recovery after photobleaching, and another used a relatively novel electrochemical method to support its FCS observation of EED (29).

Two FCS studies conducted by Jee and coworkers (27, 28) provide FCS data at increased spatial and temporal resolution. By combining superresolution microscopy and FCS, these authors were able to reduce the horizontal diameter of the observation region—the beam waist—to 50 nm, versus more typical values of approximately 800 nm (43). As the beam waist fell below approximately 100 nm, an initially unimodal distribution of transit times was resolved into two peaks. The peak corresponding to longer transit times was attributed to conventional diffusion through the waist, and the peak corresponding to shorter times was attributed to fast ballistic motions induced by catalysis.

Another study measured the diffusion of fluorescently labeled urease by single-molecule tracking with total internal reflectance fluorescence microscopy (70). This method detects molecules only while they are approximately 300 nm from the planar glass coverslip, so the enzyme molecules were confined in this layer by addition of methylcellulose to the solution. This study yielded a remarkable 300% increase in the diffusion coefficient upon addition of 1 mM urea. Interestingly, the enhanced diffusion coefficient observed here, $\sim 3 \times 10^{-13}$ m^2/s, is approximately 200-fold less than the nonenhanced value measured by FCS for urease in homogeneous solution (54). This drop in the diffusion coefficient might result from increased viscosity due to the methylcellulose. It is perhaps relevant that increasing the viscosity reduces the power required to achieve a given

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relative increment in the translational diffusion coefficient via self-propulsion (see Section 3.1, Equation 2) (17). However, such a large change in the baseline diffusion coefficient raises the possibility that the physics of diffusion in this confined setting is significantly different from that in bulk solution.

2.2. Evidence For and Against an Active Mechanism of Enhanced Diffusion

The concept that EED results from an active process has gained support from FCS and dynamic light scattering (DLS) studies indicating that catalytically active urease, catalase, and aldolase generate increased motion of passive tracer particles in solution (28, 74), with a magnitude and reaction-rate dependence similar to those of the enzymes' own enhanced diffusion. Similarly, immobilized DNA polymerase exhibits EED and generates fluid flow proportional to the catalytic rate (55). The observation of a 20% increase in the diffusion coefficient of passive tracers is particularly striking for aldolase given that its turnover rate is only approximately 5/s and that it was present at a concentration of 10 nM.

If EED results from catalysis, then enzyme inhibitors should prevent EED. Accordingly, EED has not been observed in catalase in the presence of both substrate and the inhibitor cyanide (54) or azide (46). However, at least three enzymes described as having catalysis-induced EED have also been reported to show EED in the absence of catalysis. First, Yu and coworkers (71) reported that RNA polymerase has 25% EED when catalytically active and 15% EED when substrate is provided but the required cofactor Mg^{2+} is withheld. Second, heat release had been proposed as a mechanistic requirement for EED (46); however, Illien and coworkers (25) later showed 30% EED for the enzyme aldolase, although this enzyme catalyzes an endothermic reaction (75). This led to the idea that catalysis itself might not be required for EED by aldolase, and the same authors used FCS to show that aldolase's competitive inhibitor pyrophosphate could generate nearly the same level of EED as its substrate, fructose-1,6-bisphosphate. Third, urease did not show EED in the presence of the inhibitor pyrocatechol alone at 1 mM concentration but did show attenuated EED in the presence of both pyrocatechol and substrate (43). Another study of urease found that a 1 mM concentration of the substrate urea sufficed to cause EED, while the urease inhibitor boric acid began to cause EED only at higher concentrations (approximately 100 mM) (27). It was therefore proposed that substrate and inhibitor cause EED by two different mechanisms, to account for inconsistencies among experimental results as to when or whether EED requires catalysis (27).

2.3. Negative Observations and Possible Experimental Artifacts

Significant inconsistencies have emerged in studies of aldolase across multiple techniques. Although an FCS study (25) indicated approximately 30% EED in the presence of either substrate or a competitive inhibitor, Zhang and coworkers (72) studying aldolase using DLS found no EED in the presence of either substrate or inhibitor. Günther and coworkers (22) also observed no EED when studying aldolase using a third technique, diffusion ordered spectroscopy (DOSY) nuclear magnetic resonance (NMR). It is not yet clear how to reconcile all of these results, but another study from Günther and coworkers (21) highlights potential artifacts and complexities of the widely used FCS technique.

One potential source of error in FCS is that there is always some free fluorophore, so if more protein binds to the glass over time, then the relative contribution of the fast-diffusing fluorophore to the measured diffusion coefficient increases, leading to an artifactual increase in the apparent diffusion coefficient (21). However, the results of Illien and coworkers (25) argue

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against this artifact: They found that the elevated diffusion coefficient of aldolase in the presence of substrate returned to baseline once the substrate was consumed.

Quenching of the fluorophore by substrate or product could also lead to errors, as suggested by Bai & Wolynes (6). Indeed, Günther and coworkers (21) showed that this artifact can account for apparent EED of the enzyme alkaline phosphatase (46) when the quenching substrate nitrophenyl phosphate is used because the apparent EED disappears when a nonquenching substrate is used. Given that EED in catalase has been studied by FCS, it is worth noting that its substrate, hydrogen peroxide, can also act as a quencher (21). In contrast, an FCS study of EED in urease argues against the quenching artifact by confirming that urea does not reduce the fluorescence lifetime of the fluorescent label (27).

Finally, FCS measurements are typically carried out at enzyme concentrations roughly 1,000 times lower than those of DLS and DOSY measurements (21, 72). This makes it more probable that some of the multimers have dissociated in the FCS studies. Given that binding of substrate and/or inhibitor molecules sometimes promotes dissociation (8, 68), binding could increase the measured diffusion coefficient merely by causing enzymes to dissociate into faster-diffusing subunits. In any case, because most reports of EED, for all enzymes studied, rely exclusively on FCS, it would seem important to track down the cause of the discrepancy for aldolase and/or to apply alternative experimental methods to other enzymes.

3. POTENTIAL MECHANISMS OF ENHANCED ENZYME DIFFUSION

In this section, we analyze proposed mechanistic explanations of EED. These are divided into two categories, propulsive and nonpropulsive. Before discussing specific mechanisms, we consider the thermodynamics of active self-propulsion and the possibility of a role for hydrodynamic interactions among enzymes in solution. It is worth emphasizing at the outset that, to be plausible, a mechanism must meet both qualitative and quantitative criteria. That is, the proposed mechanism must not only be physically workable but also capable of generating EED at the levels observed experimentally when realistic values of parameters, such as k_{cat} and hydrodynamic radius, are considered. For example, in principle, an enzyme's diffusion coefficient could rise due to heating of the solution by the enzyme-catalyzed reaction. However, under normal conditions, the temperature does not rise nearly enough to account for observed levels of EED (see Section 3.4.2).

3.1. A Thermodynamic Constraint on Enzyme Self-Propulsion

Varied physical mechanisms have been proposed by which chemical energy released via enzyme catalysis could lead to propulsion, thus increasing an enzyme's translational diffusion coefficient. Any propulsion mechanism necessarily leads to dissipation of chemical energy, such as by viscous drag opposing the propelled motion. Therefore, the entire class of self-propulsion mechanisms is plausible only when the catalyzed reaction provides enough power to match the unavoidable dissipation. We recently analyzed this generally applicable limit, quantitatively connecting theory with experimental data (17), as summarized in this section.

A fundamental aspect of self-propulsion mechanisms is that the enzyme is an asymmetric particle, which is considered to be propelled in a given direction within its own frame of reference. As a consequence, the enzyme's rotational diffusion causes its lab-frame propulsion direction to change stochastically, according to the rotational diffusion constant D_r. The overall translational diffusion of such a self-propelled particle results from normal Brownian motion combined with this randomly oriented self-propulsion. Using the Stokes-Einstein law, one can write an apparent diffusion coefficient as the sum of the nonenhanced translational diffusion coefficient D_t and a

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contribution from propulsion with speed v:

$$D_{\text{app}} = D_{\text{t}} + \frac{v^2}{6D_{\text{r}}}\frac{t_{\text{p}}}{t_{\text{c}}}, \qquad 1.$$

where $\frac{t_{\text{p}}}{t_{\text{c}}}$ is the mean fraction of the catalytic cycle during which the propulsive force is present. The minimal power required to achieve a certain ratio of diffusion enhancement, $R = \frac{D_{\text{app}}}{D_{\text{t}}} - 1$, is given by the Stokes drag dissipation rate (37, 48, 49, 59, 62):

$$P_{\text{req}} = 6\pi\eta a v^2 \frac{t_{\text{p}}}{t_{\text{c}}} = \frac{3}{4}\frac{R(kT)^2}{\pi\eta a^3}, \qquad 2.$$

where a is the hydrodynamic radius, and η is viscosity. Using the hydrodynamic radii and turnover numbers of enzymes reported to undergo EED allows one to estimate the free energy per reaction required to generate experimental levels of diffusion enhancement (17). The required free energies (**Table 1**) are orders of magnitude larger than those available for the slower enzymes and considerably larger than those available even for the fastest enzymes, despite the use of conservative assumptions in the power analysis (17).

The strong inverse dependence of P_{req} on hydrodynamic radius a helps explain how self-propulsion can still lead to significant enhanced diffusion for larger particles with asymmetrically disposed catalytic sites, such as so-called Janus nanoparticles (36). Intuitively, the orders-of-magnitude slower rotational diffusion of these larger particles makes the self-propulsion trajectory much less tortuous and more effective at generating net displacement from an initial location.

As part of this analysis, we considered the chemical free energy available to power propulsion. In principle, the chemical power available to drive propulsion is the product of the reaction rate and the free energy change of the reaction under the experimental conditions studied. This free energy change depends on the concentrations of reactants and products. However, global concentrations do not couple to the local processes of substrate binding, chemical reaction, and product release, so the global free energy of reaction may not be what is available to drive propulsion; it may instead be a local free energy (3) that is relevant. We used a thermodynamic cycle to estimate local free energy changes of enzyme-catalyzed reactions and found that, fortuitously, these are generally close to the standard reaction free energies (17). Experimental measurements of enzyme diffusion when the reaction is at equilibrium or running in reverse might test these ideas and shed light more generally on mechanisms of EED.

3.2. The Insignificance of Hydrodynamic Interactions

The thermodynamic analysis in Section 3.1 assumes that the enzyme molecules in solution move independently. However, EED might be amplified if the motions of each enzyme molecule could be further driven by the motions of others. Because the enzyme solutions used in FCS measurements are dilute, typically 10 nM, the average distance between two enzyme molecules is approximately 550 nm. At this range, any intermolecular forces are extremely weak. For example, the Coulombic interaction between two enzymes of charge +10, assuming the dielectric constant of water, is less than 0.1 kcal·mol^{-1}, even neglecting ionic screening. However, hydrodynamic interactions among enzyme molecules might provide a mechanism for longer-ranged enzyme–enzyme correlations. Indeed, several studies have indicated that enzyme activity, including that of aldolase, can generate fluid flows (55) or enhanced diffusion of passive tracer molecules (74), possibly via hydrodynamic interactions. In addition, Sengupta and coworkers observed no

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increase in the diffusion coefficient of fluorescently labelled, inactivated catalysis in the presence of unlabeled, active catalase (54).

A theoretical study has shown that hydrodynamic interactions are proportional to the concentration of active enzyme, and it was argued that hydrodynamic interactions among enzymes could lead to significant increments in diffusion coefficients (40). However, the numerical calculation in this study assumed an enzyme concentration of 1,000 nM, far higher than those used in FCS measurements. If the typical concentration of 10 nM is used, then the predicted enhancement in diffusion coefficient by hydrodynamic interactions comes to only 6×10^{-13} m^2/s, much less than reported values of EED. From another perspective, the flow velocity field generated by an enzyme decays no slower than r^{-2}(14, 40). Thus, in a 10 nM solution, the flow velocity generated near the surface of one enzyme (radius ~10 nm) will have decayed at least approximately 3,000-fold at the position of another enzyme (distance ~550 nm). Thus, it seems unlikely that hydrodynamic interactions contribute significantly to EED. Future measurements examining the magnitude of EED as a function of enzyme concentration might offer further insight.

3.3. Self-Propulsion Mechanisms

In this section, we consider potential mechanisms of EED that involve self-propulsion powered by the chemical energy of the enzyme-catalyzed reaction. Note that the theoretical considerations discussed above apply to all of these possibilities.

3.3.1. Mechanical swimming. By mechanical swimming, we mean the generation of propulsion by a repeated cycle of conformational changes. Microorganisms engage in mechanical swimming (14, 35), but swimming by nonmotor enzymes is not well established. The scallop theorem implies that a simple cycle of forward and reverse conformational changes cannot generate net propulsion, as any motion induced by the forward step will be undone by the reverse step (34). However, the chiral character of enzymes means that motions driven by an out-of-equilibrium chemical reaction will be directional and hysteretic (56) and thus capable of generating net propulsion (56). Furthermore, even if the conformational changes were perfectly time reversable, they could generate an increase in the translational diffusion coefficient because rotational diffusion of the enzyme during the enzymatic cycle allows the motion generated by the forward step to be along a different lab-frame axis than the motion generated by the reverse step (33). Nonetheless, mechanical swimming is an unlikely explanation for EED, based on the thermodynamic argument provided in Section 3.1 and on another study indicating that any plausible enzyme motions are too weak to generate observed levels of EED (6).

3.3.2. Pressure waves. Riedel and coworkers (46) suggested that the rapid release of heat at the catalytic step of an exothermic, enzyme-catalyzed reaction could generate a pressure wave that produces an asymmetric force on the enzyme, leading to self-propulsion. Bai & Wolynes (6) argued against this mechanism by showing that an extremely large conformational motion, along the lines of complete unfolding and refolding, would be needed. Our interpretation of the original suggestion is that the pressure wave comes not from a fast conformational change but from sudden heating at the catalytic site due to the chemical reaction. This view might avoid the concern raised by Bai & Wolynes; however, it would still be unclear how passage of a single pressure pulse through the enzyme at the speed of sound could generate a large net displacement of the enzyme. (Note that the passage of sound waves through water or air does not lead to net displacement of the molecules forming the medium.) Golestanian (20) has also provided a theoretical argument against this proposed mechanism.

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3.3.3. Bubble propulsion. The enzyme catalase has molecular oxygen as a product, and a sufficiently high density of catalase molecules on a surface can generate oxygen bubbles, leading to a propulsive force (13, 45, 50). However, generation of bubbles was, arguably, ruled out as a mechanism for self-propulsion of catalase by direct observation and by demonstration that active catalase does not increase the diffusion coefficient of nearby passive tracer molecules (54). In addition, most of the enzymes for which EED has been reported do not create a potentially gaseous product.

3.3.4. Phoretic self-propulsion. Phoretic mechanisms play an important role in self-propelled synthetic particles, such as Janus nanomotors (51), and the theory underlying these mechanisms is well-developed (42). Self-phoretic mechanisms include self-diffusiophoresis, self-electrophoresis, and self-thermophoresis, which result, respectively, from interactions of a particle with self-induced gradients of concentration, electrical potential, or temperature. These mechanisms are unlikely to explain EED because the thermodynamic limit on self-propulsion discussed above applies to phoretic self-propulsion. In addition to frictional dissipation, phoretic self-propulsion would require extra power to maintain the self-induced gradient (49), further increasing the gap between the required power and the power available from the chemical reaction. Nonetheless, it is informative to consider specific phoretic mechanisms that have been put forward.

Self-electrophoresis has been suggested as a mechanism of EED (43), but it seems unlikely. First, EED has been reported for enzymes whose substrates and products are electrically neutral, so they cannot set up an electric field. Second, self-electrophoresis would be influenced by ionic strength, but ionic strength was reported to have no influence on the apparent EED of RNA polymerase (71). Third, although self-electrophoresis has been reported for Janus nanomotors, these have spatially separated ionic flows at their cathodes and anodes and are thus well-suited to create ionic gradients and resulting electrical fields. In contrast, enzymes usually bind substrate and expel product at the same site and therefore are not as good at generating gradients.

Arguing in favor of a self-diffusiophoresis mechanism, Colberg & Kapral (10) presented simulations of enzyme-sized particles undergoing diffusiophoresis at high propulsion speeds of roughly 4 m/s. However, their study assumes a diffusion-controlled reaction with a rate constant of approximately 4×10^{10}/(M·s), which is much larger than the enzyme turnover rates listed in **Table 1**. It also assumes a concentration of substrate much higher than that in enzyme systems—the substrate was essentially a solvent—so the diffusiophoretic forces could be unrealistically large. Finally, the magnitudes of the enzyme–substrate and enzyme–product forces are arbitrary, rather than being chosen to reflect typical enzyme–substrate interactions. Thus, their model does not closely resemble an enzyme–substrate system and is of limited applicability.

Finally, Golestanian estimated the magnitude of the potential self-thermophoretic self-propulsion in catalase and found it to be negligible (20); self-thermophoresis should be even weaker for other enzymes that release less heat.

3.4. Nonpropulsive Mechanisms

In this section, we consider mechanisms of EED that do not involve self-propulsion and therefore avoid the concerns raised above regarding the inadequacy of the chemical power available.

3.4.1. Local pH change. Muddana and coworkers (43) considered whether EED could, for some enzymes at least, result from local changes in pH due to enzymatic activity. For example, ammonia, one of the products of the urease reaction, is basic and therefore can change the pH. However, measurement of the pH in the immediate vicinity of urease with a pH-sensitive fluorophore covalently bound to the enzyme revealed pH increases of up to only ~0.8, which were

judged insufficient to explain EED. Note, too, that many enzymes for which EED has been reported cannot change the pH.

3.4.2. Temperature increase. Another way for catalysis to increase the diffusion constant of an enzyme would be for the released heat (if any) to increase the temperature of the solution. However, even for an exothermic enzyme-catalyzed reaction, the increase in the bulk temperature of the solution is far too small to account for observed EED (20, 46, 60). Golestanian (20) has proposed a more refined mechanism, collective heating, which accounts for the nonequilibrium heat flow in the measuring container and for the increase in enzyme turnover with increasing temperature, and has argued that these factors can lead to a large enough temperature increase and viscosity decrease to account for EED. However, Golestanian's study used the thermal conductivity of air in its numerical analysis, and when the 30-fold higher value of water is used instead, as would seem appropriate, the predicted temperature increase no longer appears to be sufficient.

3.4.3. Changes in conformation, conformational fluctuations, and quaternary structure. If binding of substrate generates a new conformational distribution of the enzyme with a smaller mean hydrodynamic radius, then this would lead to EED. In addition, the degree of EED would correlate with the enzyme's catalytic rate, as is often observed, because increasing substrate concentration will increase both the catalytic rate and the fraction of bound enzyme. This explanation is appealing because it avoids the power requirements of propulsive mechanisms. It could also account for observations of EED induced by binding of an inhibitor or by addition of substrate to an enzyme that lacks an essential cofactor. However, experimental data compiled from the literature (73) and Brownian dynamics simulations (31) suggest that binding of a substrate or an inhibitor does not cause a large enough reduction in an enzyme's mean radius of gyration to account for observed levels of EED.

Recently, Illien and coworkers showed theoretically that the diffusion coefficient can be increased by a decrease in the thermal fluctuations of the particle radius (24, 25). They note that an enzyme may be stiffened by binding of another molecule, so addition of substrate or inhibitor could lead to concentration-dependent EED, as observed experimentally. However, the Brownian dynamics study mentioned above (31) suggests that this mechanism would not lead to observed levels of EED.

Finally, Günther and coworkers (21) have pointed out that many of the enzymes for which EED has been reported are multimeric, and that binding of substrate can lead to dissociation of multimeric enzymes. Because dissociation into smaller components would lead to an increase in the diffusion coefficient, averaged over the various multimers in solution, binding-induced dissociation could provide another nonpropulsive explanation for observations of EED. This mechanism may be particularly relevant for FCS measurements, which are typically run at low enzyme concentrations that shift the equilibrium toward dissociated states (34). Dimeric yeast hexokinase, for which EED has been observed via FCS (75), has a dissociation constant of 0.1–1.0 M, which is well above the 10-nM concentrations typically used in FCS experiments. Furthermore, three enzymes for which EED has been observed, hexokinase, urease, and acetylcholinesterase, were reported to dissociate as substrate concentrations rose above their respective k_M values (26). It is thus worth noting that, if different-sized multimers interconvert on an appropriate timescale (tens of microseconds), then this would cause an additional decay mode in the FCS autocorrelation function and thus could offer an alternative explanation for the appearance of a bimodal distribution of transit times in high-resolution FCS studies (27, 28). However, dissociation does not seem like a viable explanation for at least some observations of EED because the tetramer-to-dimer dissociation constant of aldolase is approximately 1 pM (72), so this enzyme should be quite stable as a multimer in most or all of the relevant experiments;

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the single-molecule tracking study of urease by Xu and coworkers (70) reported no influence of the concentrations of urea or enzyme on dissociation or EED. In addition, there is evidence of EED by urease at substrate concentrations where little dissociation is evident (26).

4. EXPERIMENTAL OBSERVATIONS OF ENZYME CHEMOTAXIS

Motile bacteria and eukaryotic cells have evolved mechanisms for swimming up or down gradients of dissolved compounds. This directed movement of cells is called chemotaxis, and movement up or down a gradient is termed attractive or repulsive chemotaxis, respectively. Bacteria use a molecular memory system to determine whether their recent motions have taken them up or down the gradient and adapt accordingly, while eukaryotic cells use their size to sense the direction of a gradient across the cell in real time (52). In recent years, it has been reported that enzymes can also move preferentially up or down a substrate gradient (28, 54), with apparent drift speeds up to approximately 1 μm/s for attractive enzyme chemotaxis (54) and 10 μm/s for repulsive chemotaxis (28).

Enzymes cannot meet the strictest definition of chemotaxis (66, 67) because they lack the memory and/or size required to mimic either the bacterial or eukaryotic mechanisms. Instead, the apparent directional migration of enzymes may arise from factors such as space-dependent enzyme diffusivity or diffusiophoresis, as discussed in Section 5. In this review, we follow the literature by applying the term chemotaxis to all observations of enzymes moving preferentially up or down a concentration gradient.

4.1. Attractive Chemotaxis

An early report of attractive enzyme chemotaxis used fluorescence to detect preferential displacement of RNA polymerase up a gradient of its nucleotide triphosphate substrate in a millimeter-scale device (71). Subsequent experiments have measured fluorescence intensity profiles of labeled enzymes across microfluidic flow channels fed by incoming channels containing solutions with differing compositions (**Figure 1**). For example, a left feed might contain either plain buffer or substrate in buffer, and a right feed might contain enzyme in buffer; in this case, attractive chemotaxis could manifest by a tendency of the enzyme to move to the left more in the presence of substrate than in its absence (**Figure 1*a***). Because flow in microfluidic devices is laminar, diffusion, rather than convection, dominates the relaxation of the initial nonequilibrium concentration. Such devices showed preferential diffusion of catalase and urease toward their respective substrates (12, 54) and of DNA polymerase toward either its substrate or its Mg^{2+} cofactor in the presence of substrate (55). A technically similar study of mitochondrial malate dehydrogenase and citrate synthase showed chemotaxis toward substrate in both the presence and absence of required enzyme cofactors (69). Thus, catalysis is not always necessary for enzyme chemotaxis to be observed.

A potential weakness of studies like those above is that enhanced movement of enzyme into the region of the microfluidic channel with substrate might result only from faster diffusion in the presence of substrate, rather than from any directional preference (27). Studies by Sen and coworkers address this (41, 75). Thus, when hexokinase and its substrates D-glucose and ATP were injected into the center of the channel, with plain buffer on both sides, the spread of the active enzyme across the channel was slower than when the same experiment was done with inactive enzyme. In effect, the enzyme was retained in the central channel. This focusing result implies a real tendency of the enzyme to remain close to the substrate. In addition, catalysis appears to be important in this effect because chemotaxis was attenuated when mannose, a substrate with slower turnover, was used instead of D-glucose (41, 75), and chemotaxis was absent in situations with D-glucose but no ATP, and in situations with both ATP and L-glucose present, although the latter is not a substrate (41).

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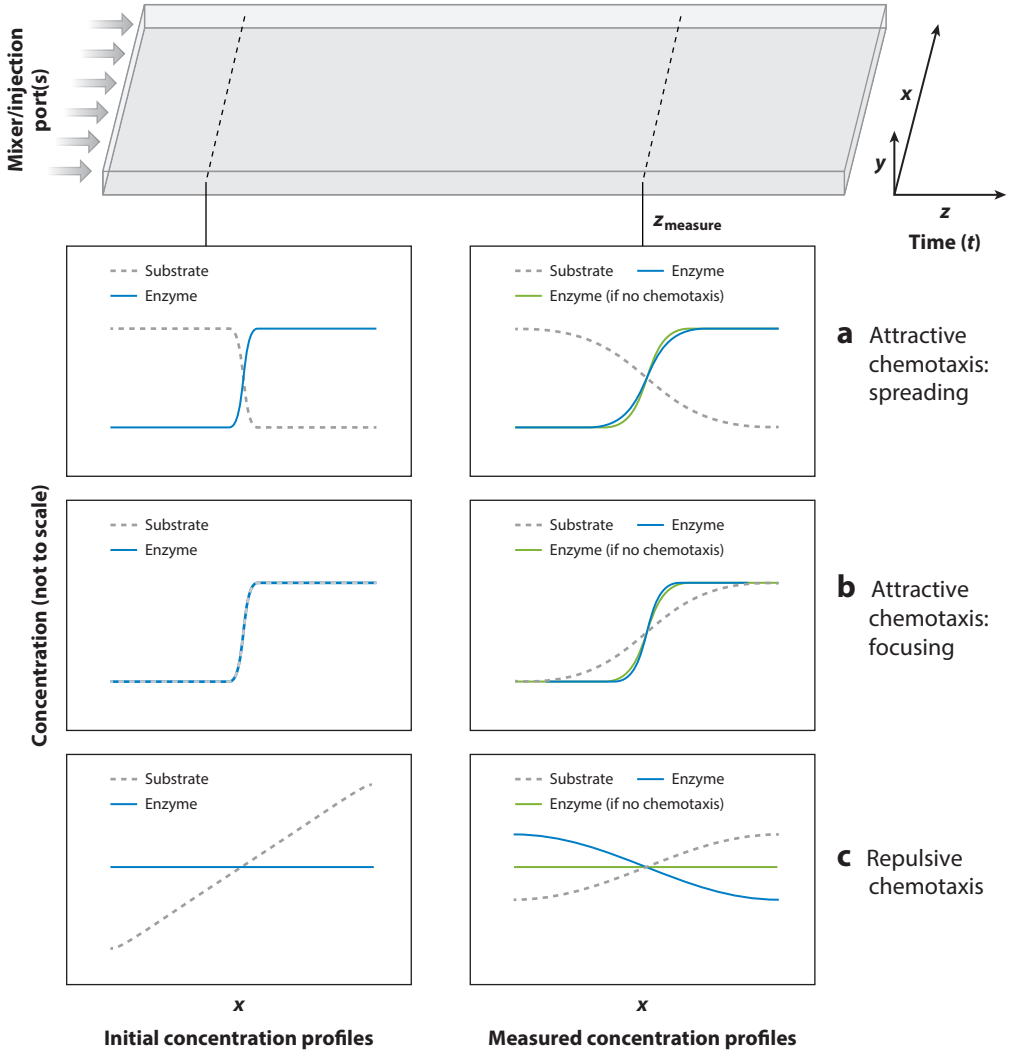

Figure 1

Generic diagram of a microfluidic chemotaxis experiment, with laminar flow from left to right, i.e., in the $+z$ direction. Concentration profiles of enzyme (*solid blue* or *green*) and substrate (*dashed gray*) across the channel (x axis) are initialized by injection ports and mixers (initial concentration profiles in the plot) and measured (measured concentration profiles in the plot) after the concentration profiles have relaxed for some time t, corresponding to distance $z_{measure}$ from the injection ports. (*a*) Concentration profiles corresponding to a transient lurch of enzyme toward substrate, suggestive of attractive chemotaxis (54). (*b*) Retention, or focusing, of enzyme in the high-substrate part of the channel, suggestive of attractive chemotaxis (75). (*c*) Repulsive chemotaxis in the context of an initially linear substrate profile (27).

4.2. Repulsive Chemotaxis

Paradoxically, Jee and coworkers (28), also using a microfluidic device, reported repulsive chemotaxis for urease and acetylcholinesterase, as an initially uniform enzyme concentration increased on the side of the channel with lower substrate concentration. Agudo-Canalejo et al. (2) have suggested that this difference, relative to prior experiments, might result from the use of a different range of substrate concentrations, but the observation was later confirmed over a

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wider concentration range (27, 41). It may also be relevant that Jee and coworkers used a different experimental design, in which the enzyme was initially at uniform concentration across the channel (**Figure 1**). In addition, it appears that Jee and coworkers' measurements (28) allowed more interaction time (distance from start of channel × channel area/fluid flow rate) than did the studies that observed attractive chemotaxis (41, 54). Thus, the experiments of Jee and coworkers might reveal something closer to the shape of the ultimate steady-state distribution of the enzyme.

4.3. The Microfluidic Method

Two important features of the microfluidic experiments are worth highlighting. First, they do not give the steady-state distribution of an enzyme in the context of a time-invariant substrate gradient, because the concentration profiles of not only the enzyme but also the substrate change as the fluid progresses along the channel. Instead, they report a transient response to a time-varying substrate gradient. Moreover, the diffusion coefficient of the substrate is usually much higher than that of the enzyme, so the concentration profile of the substrate normally relaxes more quickly than that of the enzyme. This is important because some physical mechanisms can explain transient movement of the enzyme along a substrate gradient but lead to a uniform distribution at steady state under a time-invariant substrate gradient. The microfluidic experiments may reveal the transient response but not the steady-state response. It is perhaps worth mentioning that a microfluidic setup can, in another sense, be said to have reached its steady state after enough wall-clock time has passed for its flows and concentration profiles to have stabilized. This is different from the question of whether one has taken a measurement far enough down the channel—i.e., long enough after mixing has begun—for the concentration profiles to have stabilized as a function of the mixing time. Second, when interpreting the concentration distribution of enzyme across the channel, it should be borne in mind that laminar flow has a parabolic profile, so detailed interpretation requires accounting for the details of fluid flow by simulations, as has previously been done (41).

5. POTENTIAL MECHANISMS OF ENZYME CHEMOTAXIS

Potential mechanisms of enzyme chemotaxis may be grouped into two categories. On one hand, it is posited that a force directed parallel to the substrate (or inhibitor) concentration gradient drives either attractive or repulsive chemotaxis. The other potential mechanism is based on the idea that the enzyme's diffusion coefficient depends on the substrate concentration, leading to a position-dependent diffusion coefficient when there is a substrate concentration gradient. Importantly, as emphasized by Agudo-Canalejo and coworkers (2), both categories of mechanisms can be at work in the same system. Before considering specific mechanisms, however, we present a theoretical framework for defining and analyzing proposed mechanisms of enzyme chemotaxis.

5.1. A Theoretical Framework for Mechanisms of Enzyme Chemotaxis

Enzymes translate through solvent in a Brownian manner and can thus be modeled by the Fokker-Planck (FP) equation (76). This equation describes how an initial probability distribution (i.e., concentration) in space, $p(x, t = 0)$, evolves over time, and we simplify it by considering a one-dimensional system. For example, if x represents the distance of a point from one edge of a microfluidic channel (**Figure 1**), an initial step-function concentration profile across the inlet end of the channel decays over time to a sigmoid-like distribution and approaches uniformity as the flow progresses further down the channel. The FP equation obeys conservation of probability, which requires that $\frac{\partial p}{\partial t} = -\frac{\partial}{\partial x}J$, where $J = J(x, t)$ is the probability flux. If the diffusion coefficient,

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D, is constant, then the flux can be written as $J = \mu F p - D\frac{\partial p}{\partial x}$, where the first term represents directional drift induced by a force F, and μ is the enzyme's mobility (the reciprocal of the friction coefficient).

If an enzyme diffuses at a different rate in the presence of substrate, such as by EED, then the enzyme's diffusion coefficient will depend on position, so $D = D(x)$. [The diffusion coefficient could also be a function of time, $D = D(x, t)$, because the substrate gradient in a microfluidic device decays with time (Section 4.3). However, the mathematical consequences of this complication are not considered in current literature on enzyme chemotaxis and thus are not considered in this review.] Perhaps surprisingly, merely specifying $D(x)$ does not fully determine the correct form of FP equation. This is because the flux expression depends on the character of the microscopic process that causes $D(x)$ to vary with x—an issue known as the Ito-Stratonovich dilemma, which arises in systems with multiplicative noise (16, 32, 57, 61). A range of scenarios is captured by the following expression: $J = \mu F p - \alpha \frac{\partial D(x)}{\partial x} p - D(x)\frac{\partial p}{\partial x}$, where $0 < \alpha < 1$. This leads to the following relatively general form of the FP equation:

$$\frac{\partial p}{\partial t} = -\frac{\partial}{\partial x}(\mu F p) + \alpha \frac{\partial}{\partial x}\left[p\frac{\partial D(x)}{\partial x}\right] + \frac{\partial}{\partial x}\left[D(x)\frac{\partial p}{\partial x}\right]. \quad 3.$$

In this case, $\alpha \frac{\partial D(x)}{\partial x} p$ is an additional term that could contribute to chemotaxis. In particular, if $\alpha = 1$, then the steady-state distribution of enzyme will be greater where the diffusion coefficient is lower (2, 27). In contrast, if $\alpha = 0$, then the steady-state distribution will be uniform in the absence of a force F (27, 61). The FP equation with $\alpha = 1$ is often termed the Ito form, while the FP equation with $\alpha = 0$ is termed the isothermal form (16, 57), and we use these names below.

A fundamental conclusion of this analysis is that enzyme chemotaxis cannot be mechanistically explained by merely positing a position-dependent diffusion coefficient induced by a substrate gradient. This is because the drift of the enzyme depends on α, which depends on the microscopic origin of the position dependence of $D(x)$. Nonetheless, the FP equation is a valuable framework for understanding diffusive motion, and the following section uses it to consider various possible mechanisms of enzyme chemotaxis.

5.2. Mechanisms Based on Force-Induced Drift

If a substrate gradient leads to a net force, F, on enzyme molecules in solution, then this will induce drift either up (attractive) or down (repulsive) the gradient, which could account for experimental observations of enzyme chemotaxis. This class of mechanism could explain the enzyme focusing result of Sen and coworkers (41, 75) and the evolution of an initially uniform enzyme profile into a nonuniform one in the presence of a substrate gradient (27). Force-induced drift mechanisms do not require a space-dependent diffusion coefficient, so multiplicative noise is not an issue. In this section, we consider two specific proposals for mechanisms in this class.

5.2.1. Thermodynamic force. One mechanism derives an expression for a time-averaged force on macromolecules arising from the thermodynamics of macromolecule–cosolute binding in the presence of a concentration gradient of the cosolute. Schurr and coworkers (53) included this concept in their analysis of chemotaxis of nonenzyme molecules, and good correlation with experiment was obtained for a case of enzyme chemotaxis (75). Mohajerani and coworkers (41) modified this theory for enzymes, arguing that catalysis-associated EED could magnify the effect by increasing the baseline diffusion coefficient of the enzyme and thus the speed of chemotaxis, and they reported agreement with experimental data. The fundamental picture of this model is that

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a free enzyme molecule tends to move in the direction of higher substrate concentration when it binds, whereas an enzyme–substrate complex does not have any directional preference.

A concern with this proposed mechanism is that it equates the position-averaged force on the enzyme, which is computed from the thermodynamic gradient, with the time-averaged force on the enzyme, which is the quantity relevant for chemotaxis. It is the time-averaged force that is relevant because the instantaneous velocity of an overdamped enzyme molecule is proportional to the instantaneous force, and we are interested in the mean of this velocity over time. If the free energy fell linearly with position, then the time-averaged and space-averaged forces would be equal. However, the free energy falls only at moments when the enzyme is transitioning from the unbound state to the substrate-bound state. These transitions are short lived because the binding forces are short ranged, and these brief transitions are separated by long time intervals during which the enzyme feels no directing force. The lengths of these intervals are governed by the association and dissociation rate constants. Thus, an enzyme diffusing in solution feels the binding force for only a small fraction of the time, and the time-averaged force on the enzyme is expected to be far smaller than the position-averaged force, so it is not clear how well this proposed mechanism can account for enzyme chemotaxis. This reasoning appears analogous to that in a prior study showing that the Stokes efficiency of a molecular motor is less than one and may be very low indeed when the driving potential is not linear in the spatial variable along which the motor moves (64).

Agudo-Canalejo and coworkers (1) have offered additional points of concern regarding the thermodynamic model, such as the fact that it cannot account for observations of repulsive chemotaxis (Section 4.2). The possibility of further complexities with a mechanism based on the thermodynamics of enzyme–substrate binding is suggested by a prior observation that thermodynamics alone does not determine the phoretic speeds, or even the directions, of colloid particles (3, p. 94). More broadly, determining molecular motions requires knowing more than a thermodynamic tendency, i.e., an energy gradient; one must also know how chemical coordinates and mechanical coordinates are coupled to transduce this energy (5, 63).

5.2.2. Diffusiophoresis. Diffusiophoresis causes directional drift of colloid particles up or down the concentration gradient of a cosolute (3, 30). It results from net attractive or repulsive forces between the particles and the cosolutes, leading to attractive or repulsive chemotaxis. Diffusiophoresis is similar to self-diffusiophoresis (Section 3.3.4) except that the gradient is externally imposed, instead of being self-generated by catalysis.

The particle–cosolute interactions that cause diffusiophoresis are typically nonspecific, often involve long-ranged electrostatics, and may be averaged across the surface of a relatively large colloid particle (42, 53). We are not aware of studies deriving diffusiophoretic velocities for binding of an enzyme to its substrate, but Agudo-Canalejo and coworkers (2) have argued that diffusiophoresis driven by nonspecific enzyme–substrate interactions can play a key mechanistic role in enzyme chemotaxis. However, further work is needed to assess whether the nonspecific interactions between an enzyme and its substrate are in fact capable of driving enzyme chemotaxis at observed rates via a diffusiophoretic mechanism. It is also worth noting that the magnitude and direction of nonspecific interactions may depend significantly on whether the enzyme has a bound substrate, particularly if the substrate has a nonzero net electrical charge.

5.3. Mechanisms Based on a Position-Dependent Diffusion Coefficient

If an enzyme has a higher diffusion coefficient in the presence of its substrate, then the diffusion coefficient of the enzyme will be position dependent in the presence of a substrate concentration gradient. The consequences of a position-dependent diffusion coefficient for experimental

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observations of chemotaxis are complex and case dependent (Section 5.1). For one thing, the transient effect of a position-dependent diffusion coefficient may be different from its steady-state effect. Transiently, enhanced diffusion in the presence of substrate may cause enzyme molecules introduced into the center of a microfluidic channel to diffuse preferentially to a side of the channel that has substrate, relative to a side that contains only buffer (**Figure 1**), as has previously been noted (27). This initial lurch in the direction of increasing diffusion coefficient has been termed pseudochemotaxis because it does not result from a directional preference of the diffusing particle (52), only from the fact that a nonuniform concentration distribution will relax faster where the diffusion coefficient is larger. By contrast, at steady state in the presence of a stable substrate gradient, the distribution of enzyme will depend on the value of α. If $\alpha = 0$ (isothermal form), then the steady-state concentration profile of the enzyme will be uniform, suggesting no chemotaxis. However, if $\alpha = 1$ (Ito form), then the steady-state concentration of the enzyme will be lower where the substrate is at a higher concentration (Section 5.1), matching experimental observations of repulsive chemotaxis (27, 28). In principle, the Ito form would also yield attractive chemotaxis at steady state, if the enzyme diffused more slowly in the presence of substrate, but this scenario has not been reported. Two microscopic mechanisms for a position-dependent diffusion coefficient and an Ito-form FP equation have been put forward.

One involves catalysis-driven self-propulsion of enzymes (28, 65); in effect, an enzyme gets extra propulsive kicks in the presence of substrate, leading to an increasing diffusion coefficient along a substrate gradient. This mechanism resembles that of a temperature gradient, where a particle gets more kinetic kicks in regions of higher temperature. Given that diffusion in a temperature gradient can lead to an Ito-form FP equation (32), it is likely that the self-propulsion mechanism does as well. However, the power available from enzyme-catalyzed reactions does not appear to be enough to account for the self-propulsion required by this proposed chemotaxis mechanism (Section 3.1).

The other mechanism does not involve self-propulsion, but instead involves a local equilibrium between substrate-bound and free forms of the enzyme having different diffusion coefficients. A novel derivation from Agudo-Canalejo and coworkers (2) shows that this scenario leads to the Ito-form FP equation with a position-dependent mean diffusion coefficient for the enzyme. Because this mechanism does not rely on chemical energy to power diffusive motion, it is not subject to the thermodynamic restriction of the self-propulsion mechanism.

It would be valuable to carry out more detailed and quantitative calculations that would test the ability of these mechanisms to account for the magnitudes of the effects seen experimentally. Specific questions include whether binding of substrate changes the hydrodynamic radius of urease enough to fit experiment, and whether the timescales of experiments showing repulsive chemotaxis (27) are long enough for enzyme molecules to drastically redistribute across the microfluidic channel due merely to space-dependent diffusion.

6. CONCLUSIONS

Enhanced diffusion and chemotaxis of enzymes have emerged in recent years as novel phenomena with potential implications in biology and biotechnology. They also pose intriguing puzzles whose resolution could yield new insights into molecular processes and experimental methods. Elucidation of the underlying mechanisms will require analysis of potentially subtle linkages among nonequilibrium processes spanning a range of scales. Ultimately, understanding the mechanisms should make it possible to design enzymes or other molecules to maximize these effects and put them to use. Continued work in this field promises new insights into the intricacies of molecular motions in out-of-equilibrium systems.

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FUTURE ISSUES

1. What characteristics of enzymes—e.g., structural or catalytic—correlate with EED and/or chemotaxis?
2. Are there design principles to discover and even utilize?
3. Are different mechanisms at work in different enzymes?
4. What role, if any, does the catalytic release of chemical energy play in EED and enzyme chemotaxis?
5. Why do DOSY and DLS yield diffusion results that are so different from those of FCS for aldolase?
6. Is enzyme chemotaxis purely a transient phenomenon, or can it also be observed when the enzyme concentration profile is at steady state in a stable substrate gradient?
7. What determines whether a chemotactic enzyme will undergo attractive versus repulsive enzyme chemotaxis?

DISCLOSURE STATEMENT

M.K.G. has an equity interest in, and is a cofounder and scientic advisor of, VeraChem LLC.

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LITERATURE CITED

1. Agudo-Canalejo J, Adeleke-Larodo T, Illien P, Golestanian R. 2018. Enhanced diffusion and chemotaxis at the nanoscale. *Acc. Chem. Res.* 51:2365–72
2. Agudo-Canalejo J, Illien P, Golestanian R. 2018. Phoresis and enhanced diffusion compete in enzyme chemotaxis. *Nano Lett.* 18:2711–17
3. Anderson JL. 1989. Colloid transport by interfacial forces. *Annu. Rev. Fluid Mech.* 21:61–99
4. Astumian RD. 2014. Enhanced diffusion, chemotaxis, and pumping by active enzymes: progress toward an organizing principle of molecular machines. *ACS Nano* 8:11917–24
5. Astumian RD, Mukherjee S, Warshel A. 2016. The physics and physical chemistry of molecular machines. *ChemPhysChem* 17:1719–41
6. Bai X, Wolynes PG. 2015. On the hydrodynamics of swimming enzymes. *J. Chem. Phys.* 143:165101
7. Burrows JA, Goward CR. 1992. Purification and properties of DNA polymerase from *Bacillus caldotenax*. *Biochem. J.* 287:971–77
8. Brsch M, Turina P, Eggeling C, Fries JR, Seidel CA, et al. 1998. Conformational changes of the H^+-ATPase from *Escherichia coli* upon nucleotide binding detected by single molecule fluorescence. *FEBS Lett.* 437:251–54
9. Challberg MD, Englund PT. 1979. Purification and properties of the deoxyribonucleic acid polymerase induced by vaccinia virus. *J. Biol. Chem.* 254:7812–19
10. Colberg PH, Kapral R. 2014. Ångström-scale chemically powered motors. *Europhys. Lett.* 106:30004
11. de la Torre JG, Huertas ML, Carrasco B. 2000. Calculation of hydrodynamic properties of globular proteins from their atomic-level structure. *Biophys. J.* 78:719–30

12. Dey KK, Das S, Poyton MF, Sengupta S, Butler PJ, et al. 2014. Chemotactic separation of enzymes. *ACS Nano* 8:11941–49
13. Dey KK, Sen A. 2017. Chemically propelled molecules and machines. *J. Am. Chem. Soc.* 139:7666–76
14. Elgeti J, Winkler RG, Gompper G. 2015. Physics of microswimmers—single particle motion and collective behavior: a review. *Rep. Prog. Phys.* 78:056601
15. Ey PL, Ferber E. 1977. Calf thymus alkaline phosphatase: I. Properties of the membrane-bound enzyme. *Biochim. Biophys. Acta Enzymol.* 480:403–16
16. Farago O, Grønbech-Jensen N. 2014. Fluctuation–dissipation relation for systems with spatially varying friction. *J. Stat. Phys.* 156:1093–110
17. Feng M, Gilson MK. 2019. A thermodynamic limit on the role of self-propulsion in enhanced enzyme diffusion. *Biophys. J.* 116:1898–906
18. Follmer C, Pereira FV, da Silveira NP, Carlini CR. 2004. Jack bean urease (EC 3.5.1.5) aggregation monitored by dynamic and static light scattering. *Biophys. Chem.* 111:79–87
19. Froede HC, Wilson IB. 1984. Direct determination of acetyl-enzyme intermediate in the acetylcholinesterase-catalyzed hydrolysis of acetylcholine and acetylthiocholine. *J. Biol. Chem.* 259:11010–13
20. Golestanian R. 2015. Enhanced diffusion of enzymes that catalyze exothermic reactions. *Phys. Rev. Lett.* 115:108102
21. Günther JP, Börsch M, Fischer P. 2018. Diffusion measurements of swimming enzymes with fluorescence correlation spectroscopy. *Acc. Chem. Res.* 51:1911–20
22. Günther JP, Majer G, Fischer P. 2019. Absolute diffusion measurements of active enzyme solutions by NMR. *J. Chem. Phys.* 150:124201
23. Iino R, Hasegawa R, Tabata KV, Noji H. 2009. Mechanism of inhibition by C-terminal-helices of the subunit of *Escherichia coli* FoF1-ATP synthase. *J. Biol. Chem.* 284:17457–64
24. Illien P, Adeleke-Larodo T, Golestanian R. 2017. Diffusion of an enzyme: the role of fluctuation-induced hydrodynamic coupling. *Europhys. Lett.* 119:40002
25. Illien P, Zhao X, Dey KK, Butler PJ, Sen A, Golestanian R. 2017. Exothermicity is not a necessary condition for enhanced diffusion of enzymes. *Nano Lett.* 17:4415–20
26. Jee AY, Chen K, Tlusty T, Zhao J, Granick S. 2019. Enhanced diffusion and oligomeric enzyme dissociation. *J. Am. Chem. Soc.* 141:20062–68
27. Jee AY, Cho YK, Granick S, Tlusty T. 2018. Catalytic enzymes are active matter. *PNAS* 115:E10812–21
28. Jee AY, Dutta S, Cho YK, Tlusty T, Granick S. 2018. Enzyme leaps fuel antichemotaxis. *PNAS* 115:14–18
29. Jiang L, Santiago I, Foord J. 2017. Observation of nanoimpact events of catalase on diamond ultramicroelectrodes by direct electron transfer. *Chem. Commun.* 53:8332–35
30. Khair AS. 2013. Diffusiophoresis of colloidal particles in neutral solute gradients at finite Péclet number. *J. Fluid Mech.* 731:64–94
31. Kondrat S, Popescu MN. 2019. Brownian dynamics assessment of enhanced diffusion exhibited by fluctuating-dumbbell enzymes. *Phys. Chem. Chem. Phys.* 21:18811–15
32. Landauer R. 1983. Stability and relative stability in nonlinear driven systems. *Helv. Phys. Acta* 56:847–61
33. Lauga E. 2011. Enhanced diffusion by reciprocal swimming. *Phys. Rev. Lett.* 106:178101
34. Lauga E. 2011. Life around the scallop theorem. *Soft Matter* 7:3060–65
35. Lauga E, Powers TR. 2009. The hydrodynamics of swimming microorganisms. *Rep. Prog. Phys.* 72:096601
36. Lee TC, Alarcón-Correa M, Miksch C, Hahn K, Gibbs JG, Fischer P. 2014. Self-propelling nanomotors in the presence of strong Brownian forces. *Nano Lett.* 14:2407–12
37. Lighthill MJ. 1952. On the squirming motion of nearly spherical deformable bodies through liquids at very small Reynolds numbers. *Commun. Pure Appl. Math.* 5:109–18
38. Lilie H, Bar D, Kettner K, Weininger U, Balbach J, et al. 2011. Yeast hexokinase isoenzyme ScHxk2: stability of a two-domain protein with discontinuous domains. *Protein Eng. Des. Select.* 24:79–87
39. Martin CT, Coleman JE. 1987. Kinetic analysis of T7 RNA polymerase-promoter interactions with small synthetic promoters. *Biochemistry* 26:2690–96
40. Mikhailov AS, Kapral R. 2015. Hydrodynamic collective effects of active protein machines in solution and lipid bilayers. *PNAS* 112:E3639–44

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41. Mohajerani F, Zhao X, Somasundar A, Velegol D, Sen A. 2018. A theory of enzyme chemotaxis: from experiments to modeling. *Biochemistry* 57:6256–63
42. Moran JL, Posner JD. 2017. Phoretic self-propulsion. *Annu. Rev. Fluid Mech.* 49:511–40
43. Muddana HS, Sengupta S, Mallouk TE, Sen A, Butler PJ. 2010. Substrate catalysis enhances single-enzyme diffusion. *J. Am. Chem. Soc.* 132:2110–11
44. Ortega A, Amorós D, García de la Torre J. 2011. Prediction of hydrodynamic and other solution properties of rigid proteins from atomic- and residue-level models. *Biophys. J.* 101:892–98
45. Pantarotto D, Browne WR, Feringa BL. 2008. Autonomous propulsion of carbon nanotubes powered by a multienzyme ensemble. *Chem. Commun.* 13:1533–35
46. Riedel C, Gabizon R, Wilson CAM, Hamadani K, Tsekouras K, et al. 2014. The heat released during catalytic turnover enhances the diffusion of an enzyme. *Nature* 517:227–30
47. Rieger F, Bon S, Massoulie J, Cartaud J, Benda P. 1976. *Torpedo marmorata* acetylcholinesterase; a comparison with the *Electrophorus electricus* enzyme. Molecular forms, subunits, electron microscopy, immunological relationship. *Eur. J. Biochem.* 68:513–21
48. Sabass B, Seifert U. 2010. Efficiency of surface-driven motion: Nanoswimmers beat microswimmers. *Phys. Rev. Lett.* 105:218103
49. Sabass B, Seifert U. 2012. Dynamics and efficiency of a self-propelled, diffusiophoretic swimmer. *J. Chem. Phys.* 136:064508
50. Sanchez S, Solovev AA, Mei Y, Schmidt OG. 2010. Dynamics of biocatalytic microengines mediated by variable friction control. *J. Am. Chem. Soc.* 132:13144–45
51. Santiago I. 2018. Nanoscale active matter matters: challenges and opportunities for self-propelled nanomotors. *Nano Today* 19:11–15
52. Schnitzer MJ, Block SM, Berg HC. 1990. Strategies for chemotaxis. *Biol. Chemotactic Response* 46:15–34
53. Schurr JM, Fujimoto BS, Huynh L, Chiu DT. 2013. A theory of macromolecular chemotaxis. *J. Phys. Chem. B* 117:7626–52
54. Sengupta S, Dey KK, Muddana HS, Tabouillot T, Ibele ME, et al. 2013. Enzyme molecules as nanomotors. *J. Am. Chem. Soc.* 135:1406–14
55. Sengupta S, Spiering MM, Dey KK, Duan W, Patra D, et al. 2014. DNA polymerase as a molecular motor and pump. *ACS Nano* 8:2410–18
56. Slochower DR, Gilson MK. 2018. Motor-like properties of nonmotor enzymes. *Biophys. J.* 114:2174–79
57. Sokolov I. 2010. Ito, Stratonovich, Hnggi and all the rest: the thermodynamics of interpretation. *Chem. Phys.* 375:359–63
58. Sousa R, Chung YJ, Rose JP, Wang BC. 1993. Crystal structure of bacteriophage T7 RNA polymerase at 3.3 Å resolution. *Nature* 364:593–99
59. Stone HA, Samuel AD. 1996. Propulsion of microorganisms by surface distortions. *Phys. Rev. Lett.* 77:4102
60. Tsekouras K, Riedel C, Gabizon R, Marqusee S, Pressé S, Bustamante C. 2016. Comment on "Enhanced diffusion of enzymes that catalyze exothermic reactions" by R. Golestanian. arXiv:1608.05433 [physics.bio-ph]
61. Tupper PF, Yang X. 2012. A paradox of state-dependent diffusion and how to resolve it. *Proc. R. Soc. A* 468:3864–81
62. Wang H. 2005. Chemical and mechanical efficiencies of molecular motors and implications for motor mechanisms. *J. Phys. Condens. Matt.* 17:S3997–4014
63. Wang H, Oster G. 2002. Ratchets, power strokes, and molecular motors. *Appl. Phys. A* 75:315–23
64. Wang H, Oster G. 2002. The Stokes efficiency for molecular motors and its applications. *Europhys. Lett.* 57:134–40
65. Weistuch C, Pressé S. 2018. Spatiotemporal organization of catalysts driven by enhanced diffusion. *J. Phys. Chem. B* 122:5286–90
66. Wheat PM. 2011. *Collective behavior of swimming bimetallic motors in chemical concentration gradients*. Ph.D. thesis, Arizona State Univ., Tempe
67. Wilkinson PC. 1998. Assays of leukocyte locomotion and chemotaxis. *J. Immunol. Methods* 216:139–53
68. Woodfin BM. 1967. Substrate-induced dissociation of rabbit muscle aldolase into active subunits. *Biochem. Biophys. Res. Commun.* 29:288–93

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69. Wu F, Pelster LN, Minteer SD. 2015. Krebs cycle metabolon formation: Metabolite concentration gradient enhanced compartmentation of sequential enzymes. *Chem. Commun.* 51:1244–47
70. Xu M, Ross JL, Valdez L, Sen A. 2019. Direct single molecule imaging of enhanced enzyme diffusion. *Phys. Rev. Lett.* 123:128101
71. Yu H, Jo K, Kounovsky KL, de Pablo JJ, Schwartz DC. 2009. Molecular propulsion: chemical sensing and chemotaxis of DNA driven by RNA polymerase. *J. Am. Chem. Soc.* 131:5722–23
72. Zhang Y, Armstrong MJ, Bassir Kazeruni NM, Hess H. 2018. Aldolase does not show enhanced diffusion in dynamic light scattering experiments. *Nano Lett.* 18:8025–29
73. Zhang Y, Hess H. 2019. Enhanced diffusion of catalytically active enzymes. *ACS Cent. Sci.* 5:939–48
74. Zhao X, Dey KK, Jeganathan S, Butler PJ, Córdova-Figueroa UM, Sen A. 2017. Enhanced diffusion of passive tracers in active enzyme solutions. *Nano Lett.* 17:4807–12
75. Zhao X, Palacci H, Yadav V, Spiering MM, Gilson MK, et al. 2017. Substrate-driven chemotactic assembly in an enzyme cascade. *Nat. Chem.* 10:311–17
76. Zwanzig R. 2001. *Nonequilibrium Statistical Mechanics*. Oxford, UK: Oxford Univ. Press

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